Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV

Reasoning + Answer:

<think>

Use the relativistic nonrelativistic approximation to relate momentum and kinetic energy: m_e c^2 âĪ p^2/2E for electron motion near rest.

Given kinetic energy E = 1 eV = 1.602 ÃĹ 10^-19 J, with c = 2.998 ÃĹ 10^8 m/s, then p = sqrt(2m_e E/c^2) âĪ sqrt(2 ÃĹ 9.109 ÃĹ 10^-31 ÃĹ 1.602 ÃĹ 10^-19) âĪ 5.92 ÃĹ 10^-15 kg Â· m/s.

The de Broglie wavelength is Î» = h/p, with h = 6.626 ÃĹ 10^-34 J Â· s, giving Î» âĪ (6.626 ÃĹ 10^-34)/(5.92 ÃĹ 10^-15) âĪ 1.13 ÃĹ 10^-19 m âĪ 1.23 ÃĹ 10^-7 cm.

Correct answer: G. 1.23 ÃĹ 10^-7 cm

</think>

<answer>

G. 1.23 ÃĹ 10^-7 cm

</answer>